1

(a) (2) and (3) are asserted by the network structure.

(b) $p(b, i, \neg m, g, j) \Rightarrow p(b)p(\neg m)p(i|b, \neg m)p(g|b, i, \neg m)p(j|g)$.

$p(b) = 0.9$

$p(m) = 0.1$ // $p(\neg m) = 0.9$

$p(i|b, \neg m) = 0.5$ $\Rightarrow 0.9 \times 0.9 \times 0.5 \times 0.8 \times 0.9 = \boxed{0.2916}$

$p(g|b, i, \neg m) = 0.8$

$p(j|g) = 0.9$

(c) $p(J = t \mid B = t, I = t, M = t)$.

$p(J \mid B, I, M) = \sum_g p(J|g)p(g \mid B, I, M)$.

$p(J = t \mid G = t) = 0.9$

$p(J = t \mid G = f) = 0$

$B = t, I = t, M = t$

$p(G = t \mid B = t, I = t, M = t) = 0.9$

$p(J = t \mid B = t, I = t, M = t) = 0.9 \times 0.9 + 0 \times p(G = f \mid \sim\sim) = 0.81$

$p(J = t \mid B = t, I = t, M = t) = 0.81 \rightarrow \boxed{81\%}$

2 Let B = Burglary, E = Earthquake, A = Alarm, J = John Calls, M = Mary Calls

$p(\text{Burglary} \mid \text{John Calls} = \text{True}, \text{Mary calls} = \text{True})$

Factoring $\Rightarrow p(B \mid J, m) = \alpha P(B, j, m) = \alpha \sum_e \sum_a p(B, j, m, e, a)$

$f_1(B) = P(B)$ // $f_2(E) = P(E)$ // $f_3(A, B, E) = P(A \mid B, E)$ // $f_4(J, A) = P(J \mid A)$ // $f_5(M, A) = P(m \mid A)$

$f_4(A) = P(j \mid A) \rightarrow A = t : .9,\ A = f : .05$ // $f_5(A) = P(m \mid A) \rightarrow A = t : .70,\ A = f : .01$

Remove A $\Rightarrow f_6(B, E) = \sum_a f_3(A, B, E) f_4(A) f_5(A) \Rightarrow P(A = t \mid B, E) \cdot 0.9 \cdot 0.7 + P(A = f \mid B, E) \cdot 0.05 \cdot 0.01$

$B = t, E = t : 0.95 \times 0.9 \times 0.7 + 0.05 \times 0.05 \times 0.01 = 0.598525$

$B = t, E = f : 0.94 \times 0.9 \times 0.7 + 0.06 \times 0.05 \times 0.01 = 0.59223$

$B = f, E = t : 0.29 \times 0.9 \times 0.7 + 0.71 \times 0.05 \times 0.01 = 0.183055$

$B = f, E = f : 0.001 \times 0.9 \times 0.7 + 0.999 \times 0.05 \times 0.01 = 0.0011295$

Remove E $\Rightarrow f_7(B) = \sum_e f_2(E) f_6(B, E)$ // $P(E = t) \Rightarrow 0.002$, $P(E = f) = 0.998$

$\rightarrow B = t \Rightarrow f_7(t) = 0.002 \times 0.598525 + 0.998 \times 0.59223 = 0.59224259$

$\rightarrow B = f \Rightarrow f_7(f) = 0.002 \times 0.183055 + 0.998 \times 0.0011295 = 0.00149935$

multiply prior $P(B)$

$f_8(B) = f_1(B) f_7(B)$

$\rightarrow B=t \Rightarrow f_8(t) = 0.001 \times 0.59224259 = 0.00059224$

$\rightarrow B=f \Rightarrow f_8(f) = 0.999 \times 0.00149 3351 = 0.00149185764$

Normalize $Z = 0.00059224 + 0.00149185764 = 0.00208409764$

$P(B=t \mid j, m) = 0.00059224 / 0.00208409764 \Rightarrow \underline{0.284}$ (rounded)

$P(B=f \mid j, m) = 0.0014919 / 0.00208409764 \Rightarrow \underline{0.716}$ (rounded)

Thus, Possibility of Burglary exist: $\underline{28.4\%}$. Possibility of no Burglary exist: $\underline{71.6\%}$

3. Let Cloudy $\rightarrow C$, Sprinkler $\rightarrow S$, Rain $\rightarrow R$, wetGrass $\rightarrow W$

Given: $P(C=t) = 0.5$ // $P(C=f) = 0.5$

$P(S=t \mid C=t) = 0.1$ // $P(S=t \mid C=f) = 0.5$

$P(R=t \mid C=t) = 0.8$ // $P(R=t \mid C=f) = 0.2$

$P(W=t \mid S, R) \Rightarrow$

	$R=t$	$R=f$
$S=t$	0.99	0.90
$S=f$	0.90	0.00

(a) C, R with Boolean T & F $= 2 \times 2 = \boxed{4}$

(b) $P(C \mid S, R) \propto P(C) P(S \mid C) P(R \mid C)$

$R=t \Rightarrow P(C=t \mid S=t, R=t) = \frac{4}{9}$, $P(C=f \mid S=t, R=t) = \frac{5}{9}$

$R=f \Rightarrow P(C=t \mid S=t, R=f) = 1/21$, $P(C=f \mid S=t, R=f) = 20/21$

$C=t \Rightarrow P(R=t \mid C=t, S=t, W=t) = 22/27$, $P(R=f \mid C=t, S=t, W=t) = 5/27$

$C=f \Rightarrow P(R=t \mid C=f, S=t, W=t) = 11/51$, $P(R=f \mid C=f, S=t, W=t) = 40/51$

Kernel $k(y \rightarrow y')$, From $y = (C, R) = (c, r)$:

$\Rightarrow$ resample C from $P(C \mid S=t, R=r)$

$\Rightarrow$ resample R from $P(R \mid C=c, S=t, W=t)$

Thus, Q matrix $\Rightarrow$

	(t,t)	(t,f)	(f,t)	(f,f)
(t,t)	$17/27$	$5/54$	$5/18$	0
(t,f)	$11/27$	$22/189$	0	$10/21$
(f,t)	$2/9$	0	$59/153$	$20/51$
(f,f)	0	$1/42$	$11/102$	$310/357$

(c) Q^2 is two-step transition matrix and its (x, y) entry is probability of going state $x \to$ state y in the two steps of Markov chain.

(d) Q^n as $n \to \infty$

$\Rightarrow Q^n \to 1\pi^T$ and π means the unique stationary distribution.

π is posterior $P(C, R \mid S=t, W=t)$

(e) Assuming that Q^n is available,

For evidence $S=t$ & $W=t$, Gibbs Sampling defines a Markov Chain over non-evidence variables (C, R) with Matrix Q.

For any initial distribution μ_0 over the states

$\rightarrow \mu_n = \mu_0 Q^n \xrightarrow[n \to \infty]{} \pi$, π is stationary distribution of chain.

$\pi = P(C, R \mid S=t, W=t)$.

Query is obtained by marginalizing π

$\Rightarrow P(R=t \mid S=t, W=t) = \sum_c \pi(c, R=t)$, and similar with $R=f$.

It is a way to do probabilistic inference, however this is not a practical way to do inference. In practical way, we directly simulate the Gibbs sampling instead of making Q^n.